

Sampling

Assignment 3.1

This has been already published to Canvas.

Read chapter 4.3, 4.4, and 4.5 and submit a comment.

Estimate the Standard Error by hand (you can use a calculator) from the systolic blood dataset. You have the standard deviation and the n, so it should be pretty straightforward!

Also estimate the confidence interval

Upload this by end of day Monday.

Assignment 3.2 (15 pts)

You will submit:

- 1) Answers to the questions
- 2) Your excel file with all formulas
- 2) Your R code

Raise your hand if you have any questions.

Section 3.2.1

Answer the following problems from the book (chapter 4):

2
3
8
9
13
26

Part 3.2.2

Random sampling

You are working in a small pond in New England. This pond has exactly 100 American eels (*Anguilla rostrata*). The mean total length of this population is 100-cm. We will be randomly sampling a small number ($n=12$) and estimating the mean of the population. We will also be estimating the standard error. In this case, we know the real population mean (100-cm) and thus, we can tell whether our mean estimate was close or not to the real population mean!



Figure 1: American Eel

We will be doing this pretty straightforward exercise in both Excel and program R.

On Canvas there should be 2 files. Download both of them. Use `ee1.xlsx` for excel, and `ee1.csv` for program R.

Excel

When you download and open the file you should see the following:

	A	B	C	D	E	F	G	H	I	J	K
1	Individual	Length									
2	1	109.6332763									
3	2	98.78223789									
4	3	112.8236962									
5	4	116.9490892									
6	5	117.4564559									
7	6	93.83629328									
8	7	95.73834161									
9	8	91.49019517									
10	9	91.2132606									
11	10	111.9622041									
12	11	96.25011298									
13	12	98.96260463									
14	13	86.32485398									
15	14	92.12255831									
16	15	102.6320369									
17	16	102.6240089									
18	17	76.10286812									
19	18	107.6265567									
20	19	99.36373662									
21	20	101.4255435									
22	21	91.054545									
23	22	99.7897244									
24	23	99.39345573									
25	24	106.4176948									
26	25	107.897868									
27	26	115.1522298									
28	27	101.6153997									
29	28	103.2098902									
30	29	124.9476202									
31	30	108.2792668									
32	31	78.21153348									
33	32	103.3709474									
34	33	95.12270464									
35	34	99.68174047									
36	35	100.2830909									
37	36	102.33629									
38	37	102.33629									

This file contains the size of EVERY SINGLE INDIVIDUAL in the population (100 eels). We will be sampling 15 individuals at random.

Before we do that, you should make sure I am not lying to you and check whether the mean length is actually 100.

In cell B102 you can estimate the mean **population** length by typing:

`=AVERAGE(B2:B101)`

If you get 100, you can continue, if you don't, figure out what's going on before continuing.

Let's sample!

Let's now assume we are out fishing and you catch 12 individuals (random sampling). Selecting a random sample in excel is annoying, but we will do it once. Moving forward, we will be using R.

In order to create a random sample, name column C "sample Length" and then we can sample our 12 individuals. Unfortunately, Excel does not have a great function for sampling without replacement... so, we need to add the **Data analysis add-in**:

In order to add data analysis go to File > Options > Add-Ins, then in the "Manage" dropdown, select "Excel Add-ins" and click "Go". In the available add-ins list, check the box next to "Data Analysis Add-in" and click "OK". If you absolutely cannot get this add-in, work with someone.

Now, with this data analysis add-in we can sample. Go to "Data", and on the Analyze section you should be able to see the Analysis Add-in. Click it and select "sampling".

For input range select ALL the lengths from the individuals (all 100). For number of samples select 12, and select C2:C13 as the output range.

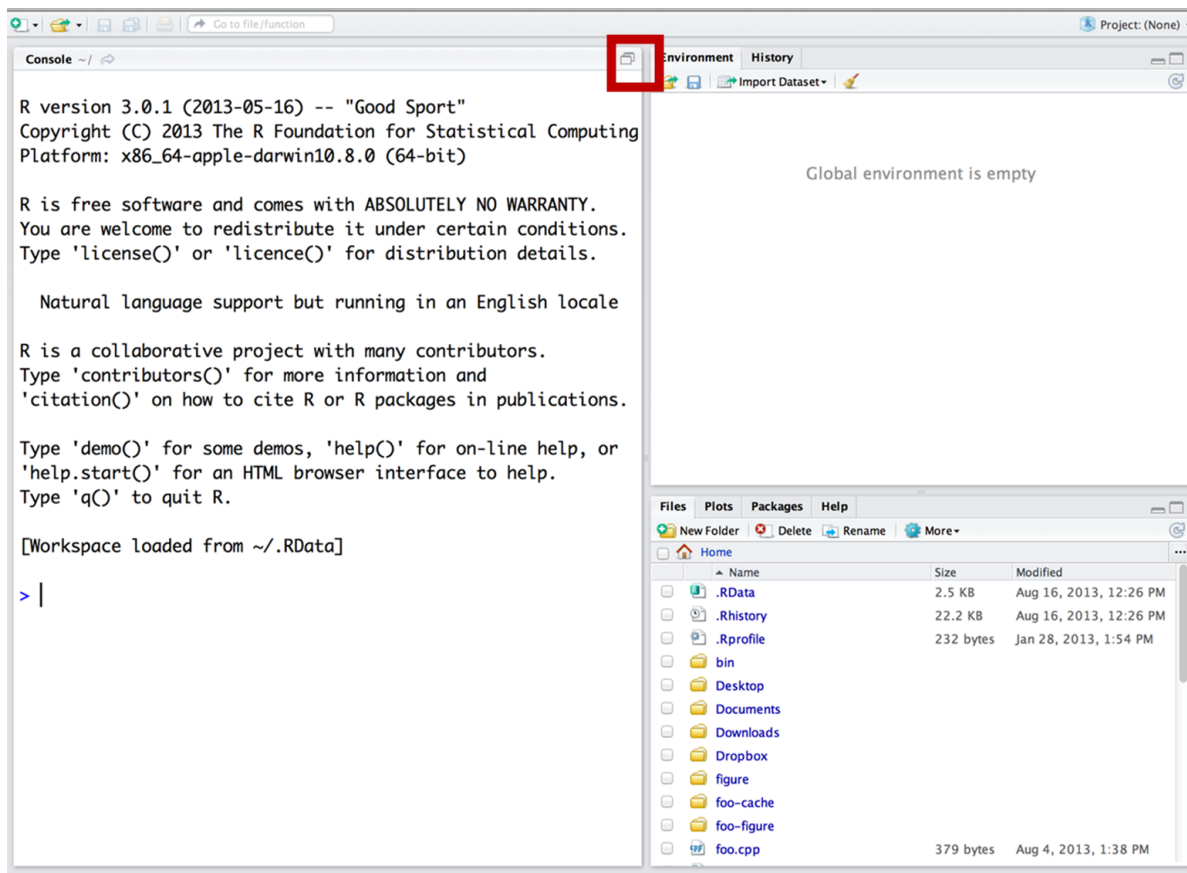
Estimates

Next, estimate the mean of your sample using the **AVERAGE** function. Estimate the Standard Deviation using **STDEV.S** and a confidence interval using the rule of thumb (see powerpoint). Let me know whether

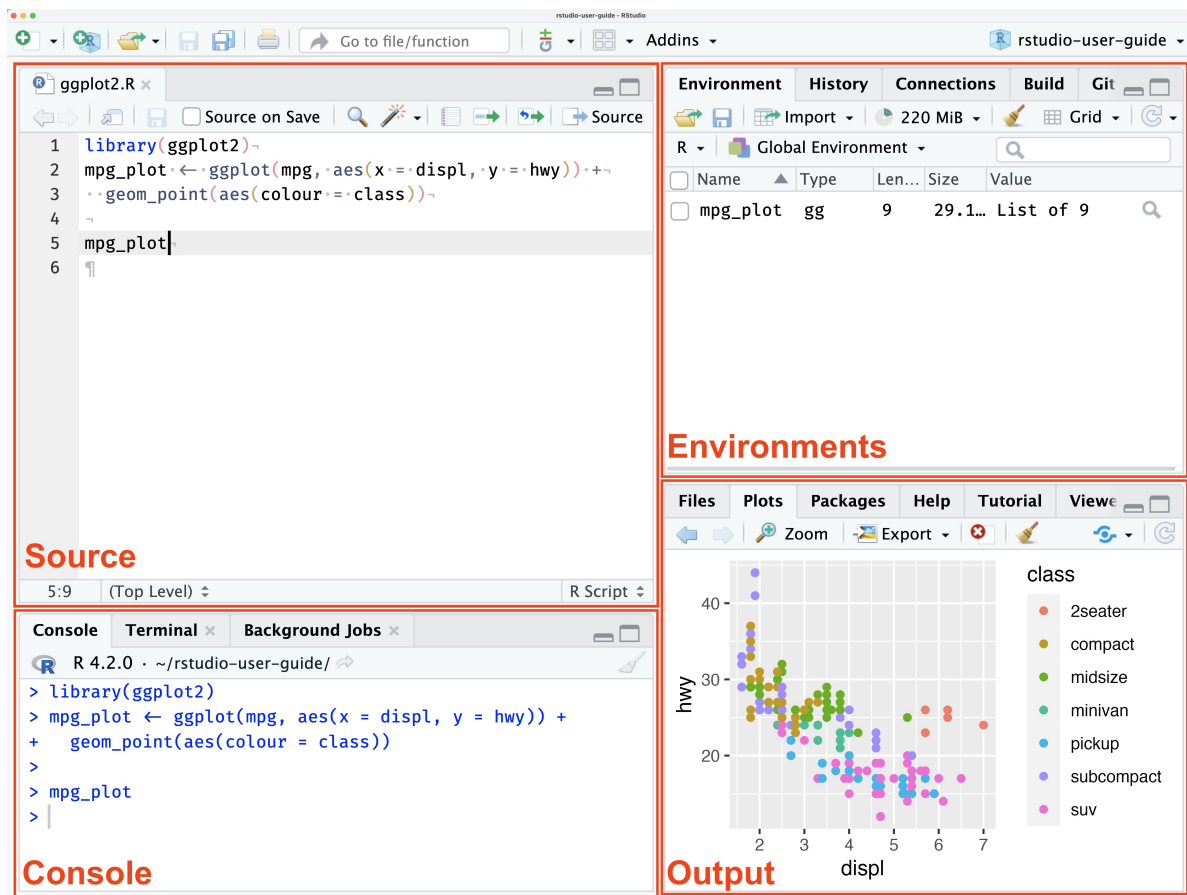
Program R

When using RStudio, you should avoid writing code in the "console". Remember this screen?

Open R-Studio and go to file > new script. If you only see three screens, make sure to press the button in the red square



and then you should look something like this:



Write all your code in the source section. Then you can run each line by pressing the **run** button or by hitting **ctrl + enter**.

You can import data of various formats into R; they include data tables in the form of **.dbf**, **.csv**, and **.xlsx** files or even spatial data such as vectors **.shp** or rasters **.nc**.

But the most common type of data files imported into R are probably **.csv** files.

You can import the dataset using the function **read.csv()**

Type this into R:

```
eel<-read.csv('eel.csv')
```

The text within the brackets, and contained in quotation marks, is the file path (directory and file name) of the file you want to import. This is my directory. Yours would most likely be different. Perhaps something like...

If you want to look at the top five rows of your head you can use the following:

```
head(eel)
```

	Individual	Length
1	1	109.63328
2	2	98.78224
3	3	112.82370
4	4	116.94909
5	5	117.45646
6	6	93.83629

We will do the same exact analysis we did in excel. First, we need to confirm the population mean:

```
mean(eel$Length)
```

```
[1] 100
```

Then, let's sample 12 individuals using:

```
sampld.eels<-sample(eel$Length,12)
```

Be aware than your sample will be different than your excel sample.

You need to estimate the mean using `mean`, the standard deviation using `sd`, and the confidence interval using the following equation:

$$CI = mean \pm 2 \times std.err$$

So, to estimate the mean use:

```
std.dev<-mean(sampld.eels)
```

And to estimate the standard error you use:

$$std.err = st.dev \div \sqrt{n}$$

And remember that n is 12.

Answer:

Did the confidence interval include the real mean of 100?

Upload the excel file and the R file . Make sure your R file has the steps to estimate the mean, standard deviation, and the confidence intervals.